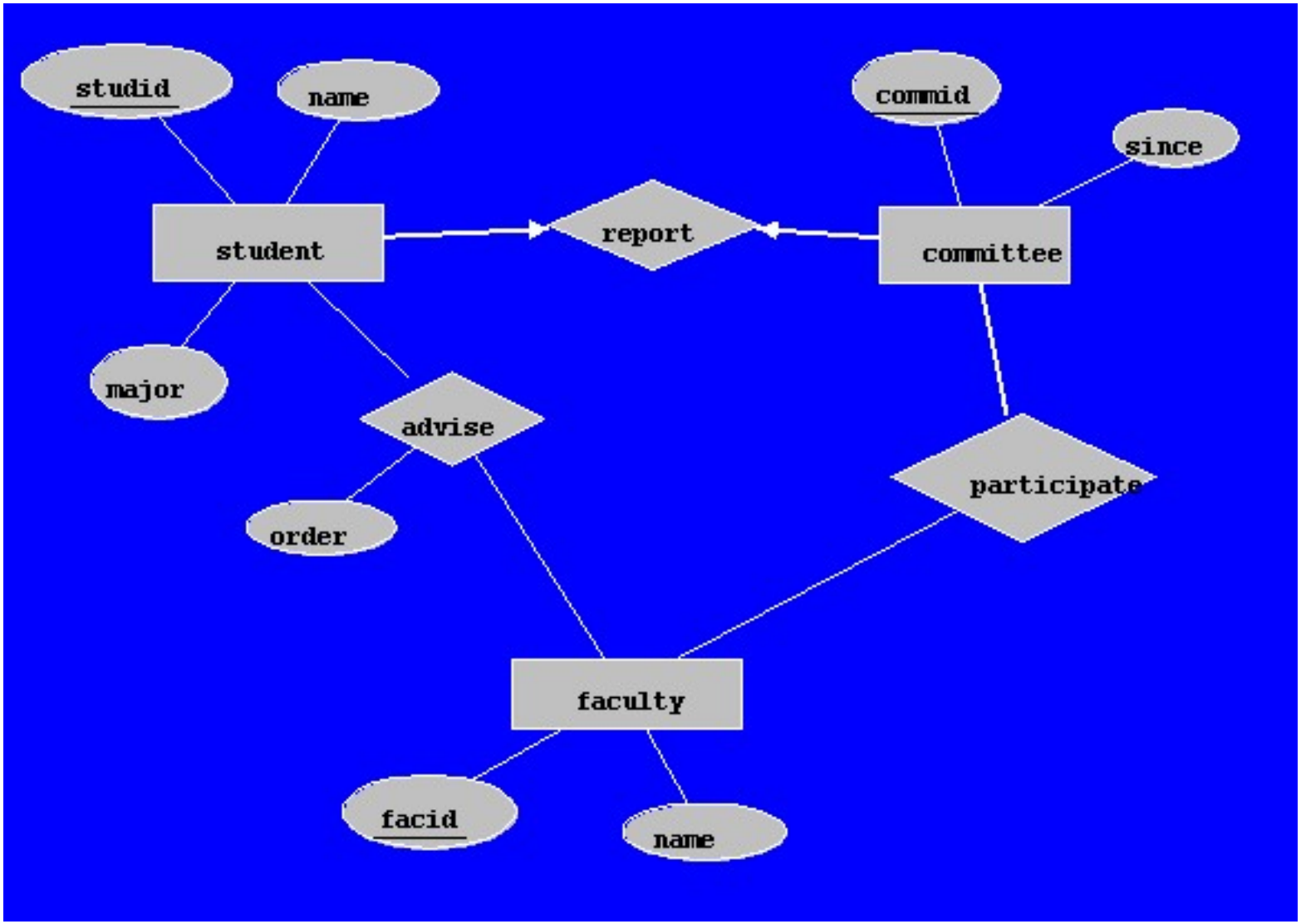
Database Design

- Goal: specification of database schema
- Methodology:
  - Use *E-R model* to get a high-level graphical view of essential components of enterprise and how they are related
  - Convert E-R diagram to DDL
- *E-R Model*: enterprise is viewed as a set of
  - *Entities*
  - *Relationships* among entities

A Sample ER Diagram



Mapping natural language

ER components can be equated to parts of speech, as Peter Chen did. This shows how an ER Diagram compares to a grammar diagram:

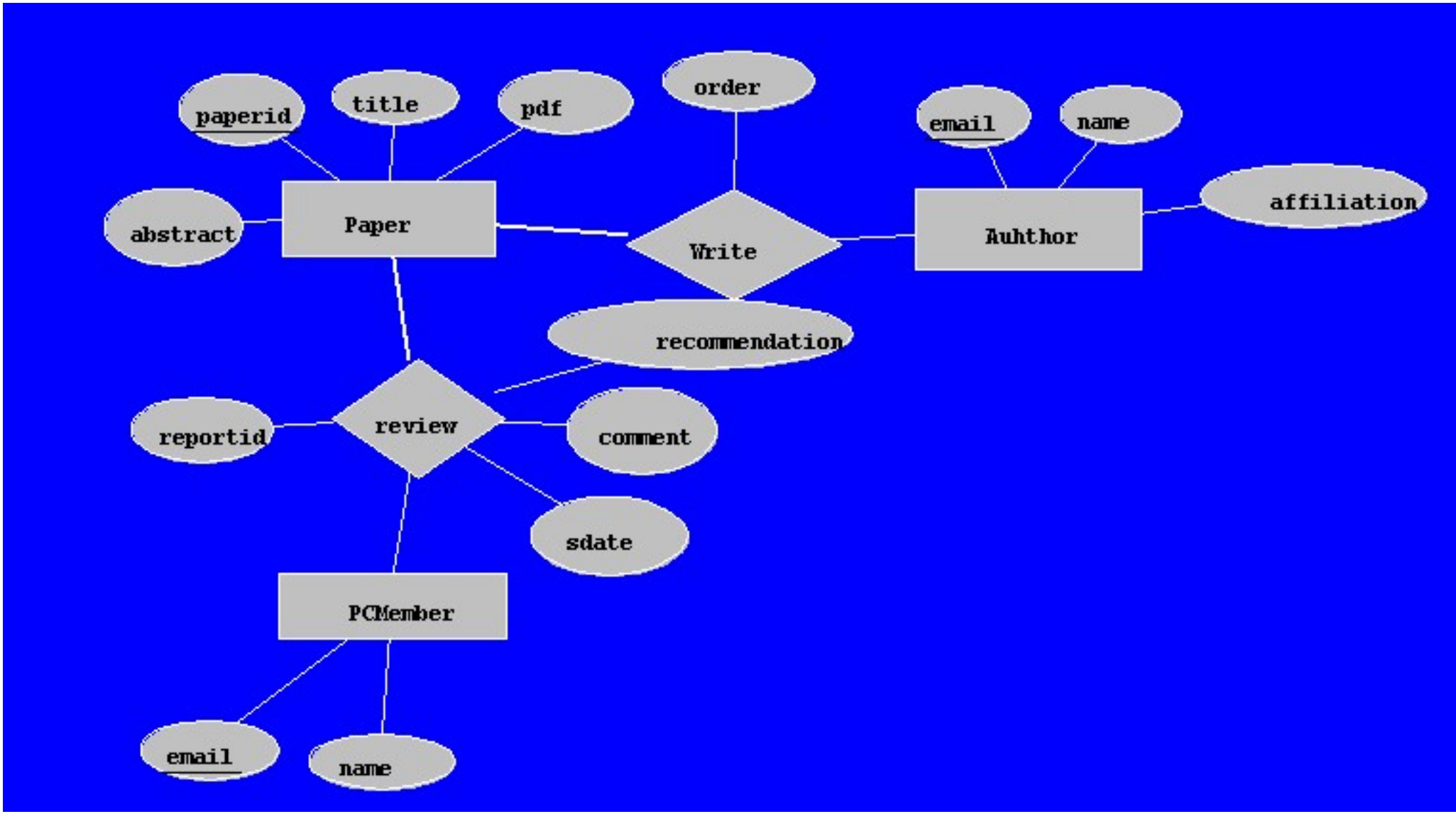
- **Common noun:** Entity type. Example: student.
- **Proper noun:** Entity. Example: Sally Smith.
- **Verb:** Relationship type. Example: Enrolls. (Such as in a course, which would be another entity type.)
- **Adjective:** Attribute for entity. Example: sophomore.
- **Adverb:** Attribute for relationship. Example: digitally.

https://www.lucidchart.com/pages/er-diagrams

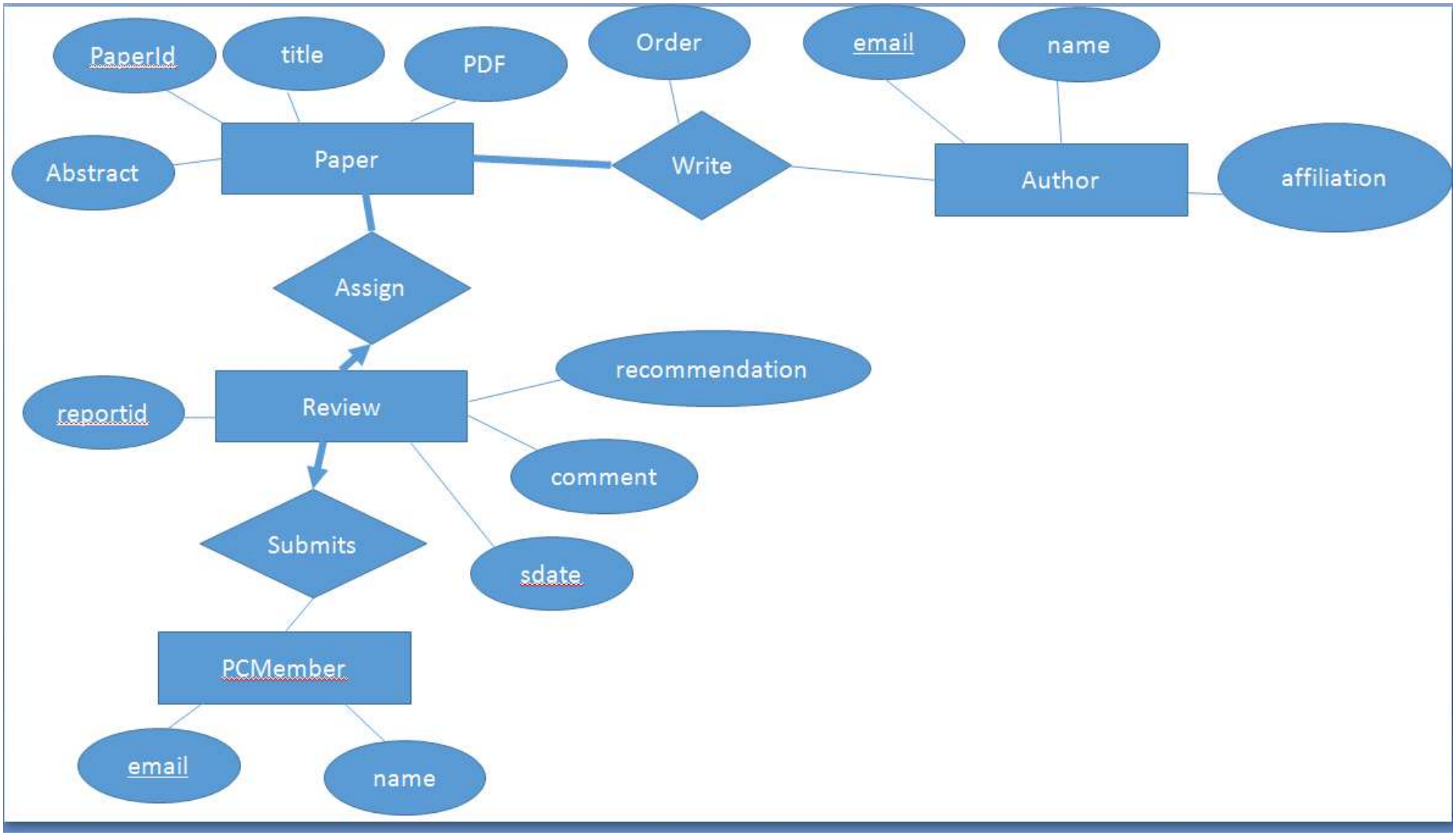
ER case study 1

Consider the design of the following database system for managing a conference X: a collection of papers are submitted to X, each of which has a unique paper IDs, a list of authors (names, affiliations, emails) in the order of contribution significance, title, abstract, and a PDF file for its content. The conference has a list of program committee (PC) members to review the papers. To ensure review quality, each paper is assigned to 3 PC members for review. To avoid overloading, each PC member is assigned with at most 5 papers, assuming that there are enough PC members. Each review report consists of a report ID, a description of review comment, a final recommendation (accept, reject), and the date the review report is submitted. A PC member can submit at most one review report for the paper that is assigned to him/her.

ER Diagram

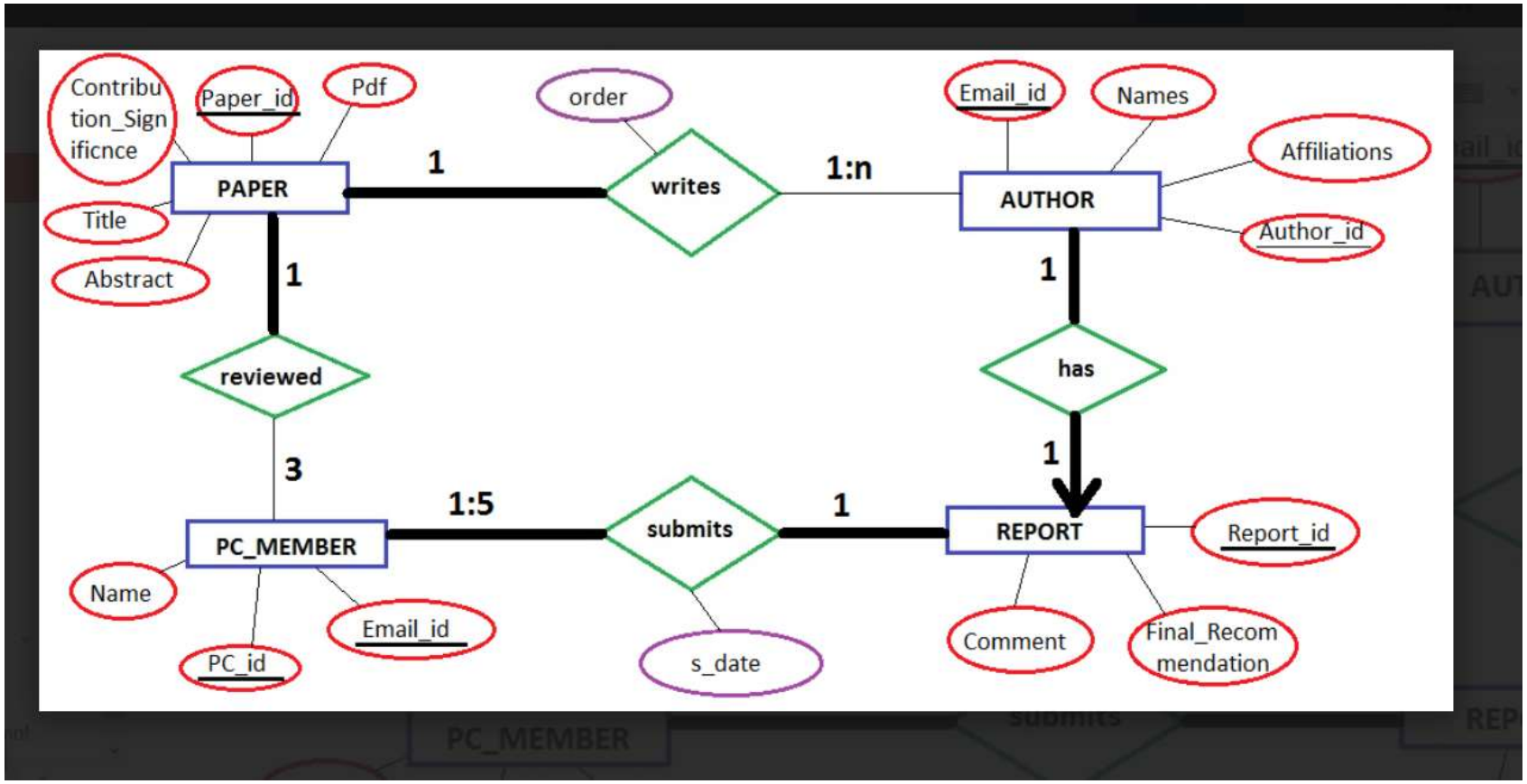

paperid
title
pdf
order
email
name
abstract
Paper
Write
Auhthor
affiliation
recommendation
review
reportid
comment
sdate
PCMember
email
name

How about modeling 'Review' as an entity type?



Credit of Renuka Gangireddy

Another notation, it correct yet?



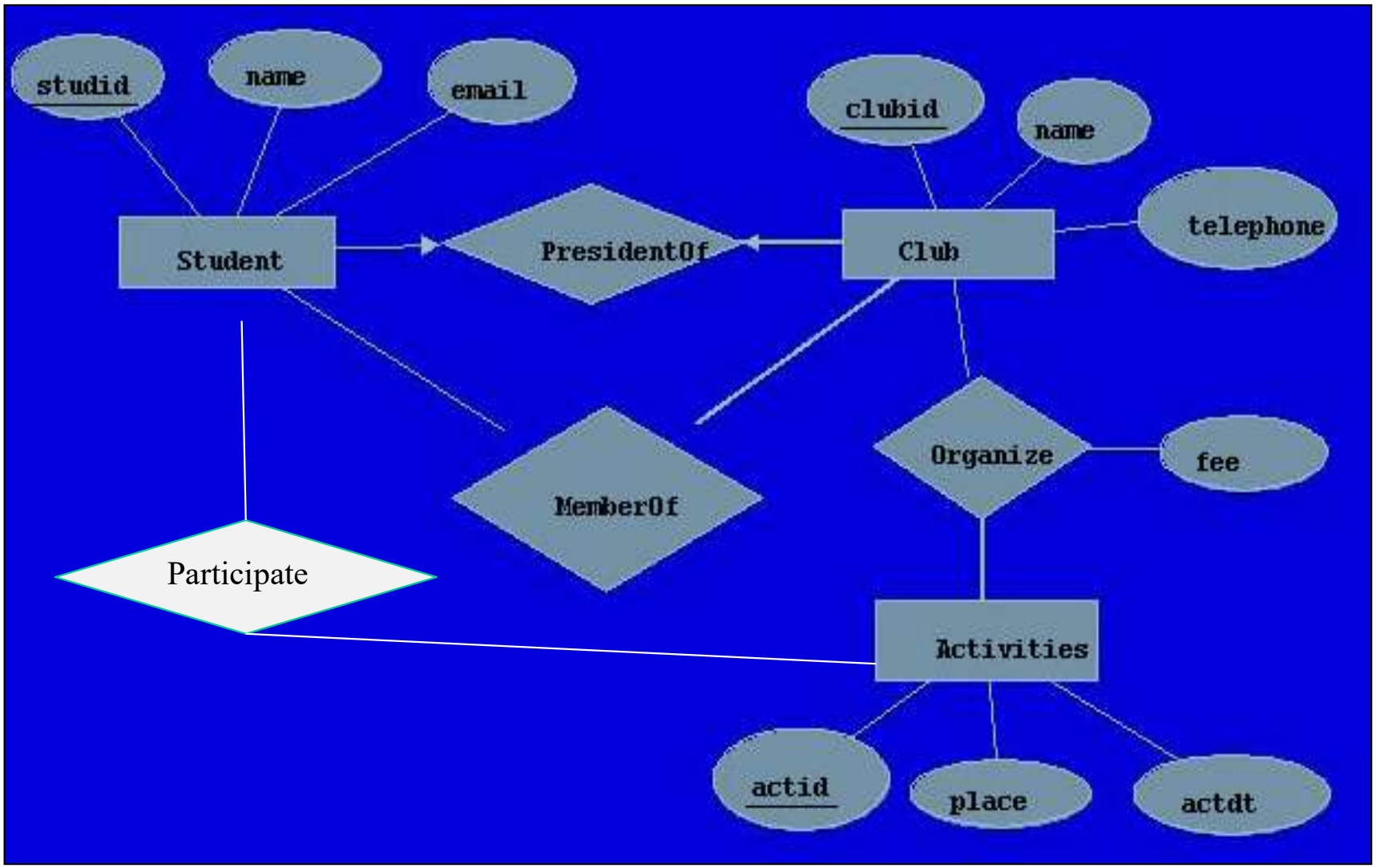
Credit of Yukti Dhiman

ER case study 2

Suppose you are asked to design a club database system based on the following information.

Each student has a unique student id, a name, and an email; each club has a unique club id, a name, a contact telephone number, and has exactly one student as its president. Each student can serve as a president in at most one of the clubs, although he/she can be the members of several clubs. Clubs organize activities and students can participate in any of them. Each activity is described by a unique activity id, a place, a date, a time and those clubs that organize it. If an activity is organized by more than one club, different clubs might contribute different activity fees.

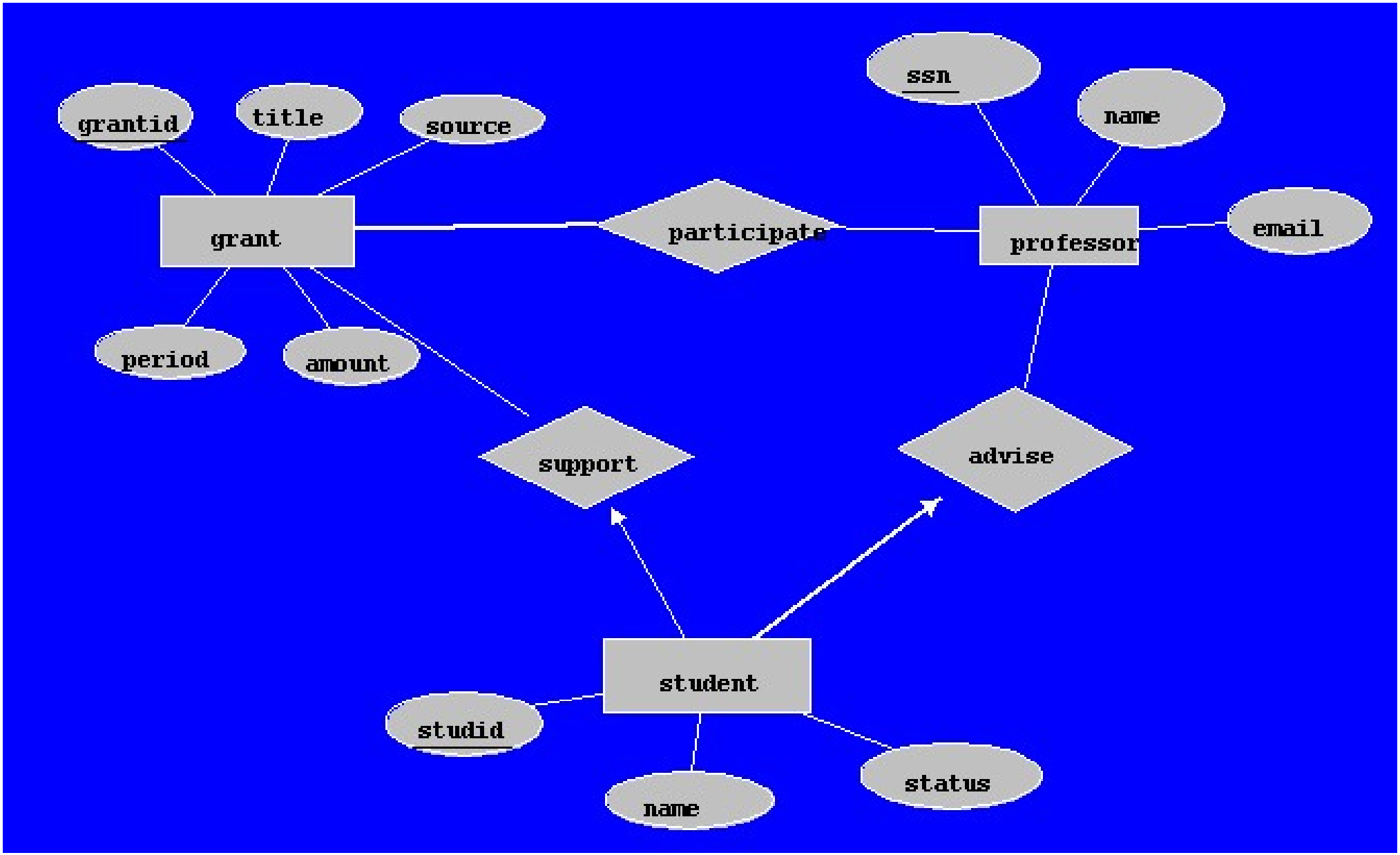
Reference solution



ER case study 3

Consider the design of a database for the management of grants. Each grant is identified by a unique grant ID, a title, the funding source of the grant, the period (starting date and ending date), and the amount of grant. Each grant might be participated by several professors and each professor might also participate in several grants. Each professor is identified by a unique SSN, name, and email address. In addition, several graduate students might be supported by a grant as GRAs, although each student can be supported by at most one grant. Each graduate student has exactly one professor as his/her advisor.

Reference solution



ER case study 4

Consider the design of the following database system: each PhD student has exactly one dissertation committee which consists of 4-5 faculty, and each committee is for exactly one student. Each student has an ordered list of advisors including the primary advisor followed by 0 or more secondary advisors. Each student has a unique studid, a name, and a major. Each committee has a unique committee id, and the date the committee is formed. Each faculty has a unique facid and a name. Each faculty can participate in multiple committees and be the advisors (either primary or secondary) of several students.

Reference solution

